

Djordje Miladinovic

Curriculum Vitæ

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Summary

- **My profile:** AI/ML research engineer | PhD in Computer Science (AI/ML-focused) | Swiss resident | Serbian citizen. 9+ years in machine learning research | 13+ years in software engineering | 4+ years in project leading.
- **My work:** Leading large-scale machine learning projects from concept to working system. This includes designing and training deep neural networks to process typically high-dimensional and multi-modal data.

Experience



Machine Learning Researcher & Engineer, *GSK-AI Department*

Oct'21-Present

- Leading GSK efforts to apply machine learning tools to massive amounts of omics data (perturbational & observational). My mission is to build a computational platform for fast, precise, and interpretable identification of disease-relevant genetic targets.



Machine Learning Researcher, *ETH Zürich*

Feb'17-Jun'21

- **Sleep-learning web platform:** Led a team of researchers to develop a web-based large-scale machine learning platform for sleep pattern recognition from EEG/EMG signals – *over 30K jobs processed to this date*.
- **SDN neural networks:** Co-developed a new type of neural network for realistic image synthesis. I then used it to *develop a state-of-the-art variational autoencoder* for image modeling.
- **Zurich Exhalomics:** In an interdisciplinary collaboration, co-developed a machine learning algorithm for interpretable identification of causal relations between metabolites and sleep stages (relating metabolism to sleep).



Machine Learning Researcher (Visitor), *Max Planck Institute for Intelligent Systems* ***Sep'18-Dec'18***

- **Causal disentanglement framework:** Co-developed a neural architecture for evaluating disentangled representations.



Machine Learning Researcher & Engineer (Intern), *Logitech Europe S.A.*

Sep'16-Feb'17

- Developed a machine learning platform to target potential premium users to reduce churn rate.



Machine Learning Researcher (Visitor), *Disney Research Studios, Zurich*

Jan'16-Sep'16

- Developed a software platform for detecting mechanical failures in Disney's humanoid robots based on IMU sensor readings. The platform uses novel machine learning algorithm to assess robot degradation in a human-like way.

Education



PhD in Computer Science – Machine Learning, *ETH Zürich*

Sep'17-Jun'21

- **Thesis:** "On training Deep Generative Models with Latent Variables" – [Prof. Joachim M. Buhmann](#).
- **Research areas:** Generative image and text modeling | variational autoencoders | representation learning | deep learning | computer g | natural language processing | machine learning for biology.



MS in Computer Science – Machine Learning, *ETH Zürich*

Sep'13-Jun'16

- **Thesis:** "Perceptual Analysis Framework for Discovering Anomalies in Humanoid Motions" – [Prof. Otmar Hilliges](#).
- **Focus:** Machine learning | software engineering | operating systems | distributed computing.



BS in Computer Science – Software Engineering, *ETF Belgrade*

Sep'09-Sep'13

- **Thesis:** "Java-based Interactive Software Simulator of a CISC Processor" – [Prof. Zaharije Radivojević](#).
- **Focus:** Software engineering and design | algorithms | data structures | operating systems | hardware design.

Programming Languages

Advanced Python | PyTorch | Bash | $\LaTeX$
Intermediate C/C++ | Java | Matlab | SQL | HTML | CSS
Basic Tensorflow | Torch | Node.js | C# | Jekyll | Javascript | VHDL

Selected Publications

- [1] **Đorđe Miladinović**, Kumar Shridhar, Kushal Jain, Max B Paulus, Joachim M Buhmann, Carl Allen
Learning to Drop Out: An Adversarial Approach to Training Sequence VAEs
Advances in Neural Information Processing Systems, **NeurIPS 2022**
- [2] **Đorđe Miladinović**, Aleksandar Stanić, Stefan Bauer, Jürgen Schmidhuber & Joachim M. Buhmann
Spatial Dependency Networks: Neural Layers for Improved Generative Image Modeling
International Conference on Learning Representations, **ICLR 2021**
- [3] Raphael Suter, **Đorđe Miladinović**, Stefan Bauer & Bernhard Schölkopf
Robustly Disentangled Causal Mechanisms: Validating Deep Representations for Interventional Robustness
International Conference on Machine Learning, **ICML 2019**
- [4] **Đorđe Miladinović** et al.
SPINDLE: End-to-end Learning from EEG/EMG to Extrapolate Animal Sleep Scoring Across Experimental Settings, Labs and Species, **PloS Computational Biology 2019**

See the complete list at [Google Scholar](#)

Academic Activities

Dec'19 Co-organized the [Disentanglement Challenge at NeurIPS 2019](#).
2018-Present [Associated research fellow](#) of [Max Planck Institute for Intelligent Systems](#).
2017-2021 Reviewed papers for NeurIPS, ICML and ICLR.
2017-2020 Taught [Advanced Machine Learning](#) and [Statistical Learning Theory](#) courses at [ETH Zürich](#).
2017-2020 Supervised more than [10 MS students](#).

Languages

Serbian (Native), English (Fluent), German (Intermediate), Spanish (Beginner)

Hobbies

Waterpolo | Swimming | Skiing and snowboarding | Tennis | Cinematography | Reading